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Three-Pillar In Vivo Blood-Brain Barrier (BBB) Pharmacokinetic Resolution

Reconciling Low Micromolar In Vivo Circulating SCFAs (1–10 μM) with High Millimolar In Vitro HDAC IC_{50} (500–2,000 μM)

PILLAR 1

BAM Vascular Sensor Relay

Perivascular Border Interface

CEREBRAL CAPILLARY LUMEN — SCFAs (1–10 μM)

Border Macrophage (BAM / Mrc1$^+$ Lyve1$^+$) — TGF-β / IL-10 — Parenchymal Microglia

- **Anatomical Positioning:** BAMs populate leptomeningeal, dural, and Virchow-Robin perivascular spaces.
- **High MCT1 Transporter Expression:** Blood-facing interface senses micromolar SCFAs without full parenchymal entry.
- **Secondary Paracrine Cascade:** Transduces homeostatic cues (TGF-β1, IL-10) across glia limitans to resting microglia.
- **Resolves In Vitro Disconnect:** Explains how peripheral depletion alters microglia despite 1–10 μM in vivo levels.

PILLAR 2

Central Acetate / ACSS2

Nuclear HAT Epigenetic Flux

Blood Acetate (100–300 μM) — MCT1 (Km ~2 mM) — Microglial Cytosol (Acss2 = 3.34 CPM)

Nucleus: ACSS2

Acetate → Acetyl-CoA → HATs

Open Chromatin Poising (88.9%)

- **Abundant Circulating Acetate:** Unlike butyrate/propionate (<10 μM), acetate reaches 100–300 μM in systemic serum.
- **Efficient MCT1 Brain Penetration:** Crosses endothelial BBB via robust monocarboxylate transporter MCT1 expression.
- **Intranuclear HAT Substrate Capture:** Microglial ACSS2 captures acetate directly to synthesize acetyl-CoA inside the nucleus.
- **Epigenetic Poising Mechanism:** Fuels histone acetyltransferases directly, sustaining poised chromatin without HDAC IC_{50}.

PILLAR 3

Enteric-Vagal Reflex

Neuro-Humoral Circuit

Gut Mucosa — SCFA Sensing (FFAR2/3) — Enteroendocrine — Vagus Nerve (Nodose) — Brainstem (NTS)

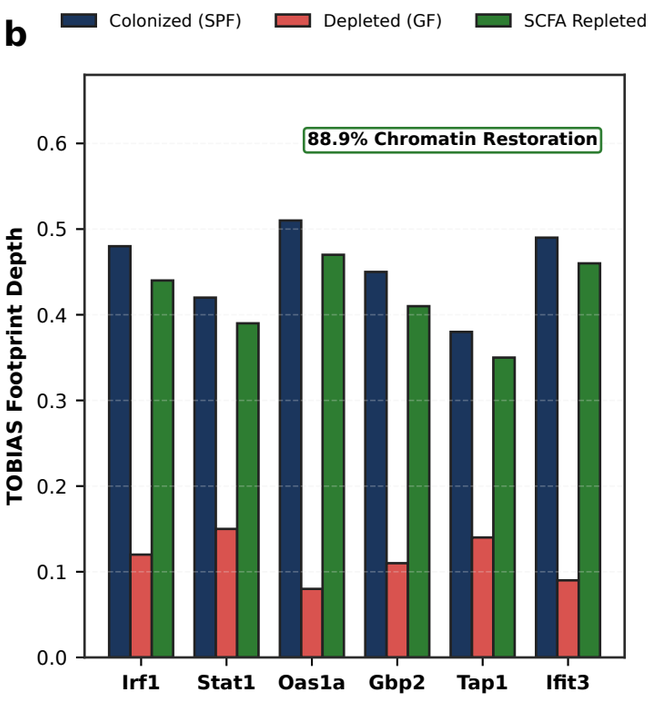
Central Cholinergic & Noradrenergic Projections → Microglial Quiescence

- **Mucosal Enteric Sensing:** Intestinal luminal SCFAs stimulate enteroendocrine L-cells and vagal terminals.
- **Afferent Vagal Conduit:** Signals ascend via nodose ganglion to nucleus tractus solitarius (NTS) in brainstem.
- **Ascending Neuro-Immune Projections:** Locus coeruleus noradrenergic and cholinergic relays modulate parenchymal microglia.
- **Non-Vascular Communication:** Enables immediate peripheral calibration without requiring direct BBB solute crossing.

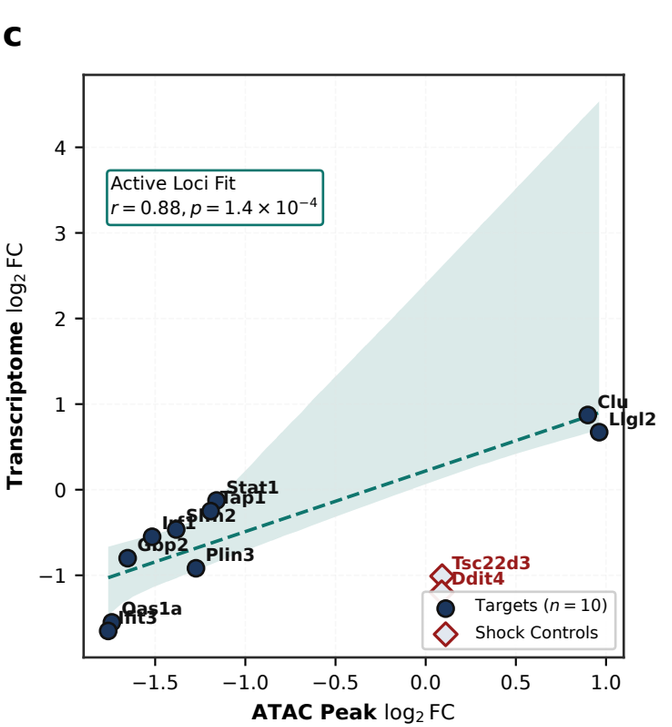
Resolution of the In Vivo Pharmacokinetic Paradox:

Direct in vitro competitive HDAC inhibition requires millimolar concentrations (IC_{50} 500–2,000 μM), but circulating gut SCFAs are only 1–10 μM in vivo.
Microglial homeostasis is sustained via BAM vascular sensing, central acetate capture by ACSS2 for HAT poising, and vagal sensory afferent circuits.

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